

Riley Aaron Gabrielson

Staff Software Engineer

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WORK EXPERIENCE

TCN · *Staff Software Engineer*

August 2021 — Present

- Founded and lead the Business Intelligence team, owning the analytics product end to end — query authoring, dashboards, scheduled report delivery, and export — and growing it from a single feature into a platform other product teams build on.
- Designed and built **Insight Builder**, a node-graph query editor where non-technical users compose joins, derivations, filters, and aggregations on a visual canvas that compiles to executable SQL, paired with a grammar-aware guided expression editor offering contextual function, parameter, and literal suggestions.
- Architected the embeddable dashboard platform: packaged BI as a versioned npm library and a shadow-DOM-isolated plugin bundle, letting any team drop live analytics into their own application without a BI deploy or CSS collisions.
- Co-architected the company's micro-frontend platform — independently released runtime services for OIDC authentication across regions, routing, plugin loading, rendering, and logging — and wrote the CI tooling that enforces service interface types and blocks breaking changes across consumers.
- Shipped a client-side analytics engine on DuckDB-WASM, querying Parquet over HTTPS in the browser to cut dashboard interaction latency and take repeat aggregation load off the backend.
- Merged 1,700+ changes and reviewed 1,000+ more from teammates, setting the frontend conventions, presenter-pattern architecture, and keyboard-accessibility standards the organization builds against.

Bad Crow Games · *Technical Lead & Game Designer*

February 2019 — August 2021

- Built the studio's websites and wrote automation scripts that took repetitive production work off the team.
- Designed and built a solo and cooperative system in which players face a rules-driven AI opponent, extending competitive titles to single-player and full-table co-op without a human adversary.
- Designed game systems and mechanics for the studio's titles.

EDUCATION

Brigham Young University · *B.S. Applied and Computational Mathematics*

September 2018 — May 2022

- Emphasis on numerical methods, optimization, and algorithm design; worked as a technical lead in industry throughout the degree.

TECHNOLOGIES & LANGUAGES

Languages TypeScript, JavaScript, Python, SQL, PRQL, GDScript

Frontend React, micro-frontends, Web Components & Shadow DOM, design systems, Storybook, Vite

Backend API design, gRPC & Protocol Buffers, REST, Node.js, DuckDB

Practice Full-stack architecture, plugin systems, monorepos, semantic release, CI/CD, technical leadership